SOCIOL 723: Social Statistics II

Week 3 — Maximum Likelihood: Theory

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Today

Why Likelihood

Two Ways to Justify an Estimator

Design / agnostic

- Start with the CEF. β is defined as the best linear approximation to it, whatever the world looks like.
- Assumes almost nothing about how Y was generated.
- This was Weeks 1–2.

Model-based

- Write down a full *probability model* for how Y was generated, with unknown parameters.
- Estimate the parameters that make the observed data most probable.
- This is maximum likelihood, and it is Weeks 3–4.

Neither is "the right one"

The design-based view is safer for causal estimands. The model-based view buys you things the agnostic view cannot deliver: predictions on the right support, structural parameters, latent variables, and outcomes that are not real numbers. Be pragmatic, and be explicit about which you are using.

OLS Is Already a Maximum Likelihood Estimator

Suppose we assume the full model

$$Y_i = X_i'\boldsymbol{\beta} + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2) \quad \Longrightarrow \quad Y_i \sim \mathcal{N}(X_i'\boldsymbol{\beta},\ \sigma^2).$$

Then the probability of the observed sample, as a function of the parameters, is

$$\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - X_i'\boldsymbol{\beta})^2}{2\sigma^2}\right).$$

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Taking logs, everything except $-\frac{1}{2\sigma^2}\sum_i (Y_i - X_i'\boldsymbol{\beta})^2$ is free of $\boldsymbol{\beta}$. Maximizing over $\boldsymbol{\beta}$ is exactly minimizing the sum of squared residuals.

So OLS under normality *is* the MLE. What we do today is take the same idea and apply it when the outcome is not normal.

Why We Need the Generalization

The normal linear model assumes $Y_i \in \mathbb{R}$ with constant variance. Sociological outcomes are often not like that.

- **Binary**: employed or not, voted or not, married or not. A linear model can predict probabilities below 0 and above 1.
- **Counts**: number of children, arrests, publications. Non-negative integers, right-skewed, variance tied to the mean.
- **Categorical**: occupation, party, religious affiliation. There is no meaningful numeric coding of {professional, service, manual}.
- **Ordered**: strongly agree . . . strongly disagree. The gaps between categories are not known.
- **Durations, censored, truncated**: time to first birth; income top-coded at \$200,000.

Maximum likelihood handles all of these, and is the engine behind almost every model in the sociological literature.

Where We Are Going

1. **The likelihood**: what it is, and the crucial distinction between probability and likelihood.
2. **Finding the maximum**: the score function, the first-order condition, and worked examples you can do on paper.
3. **Information**: how the *curvature* of the log-likelihood tells you how precise your estimate is.
4. **Properties**: consistency, asymptotic normality, efficiency, invariance — and what can go wrong.
5. **Testing**: the likelihood ratio, Wald, and score tests, which are three ways of measuring the same distance.

As in Weeks 1–2, everything examinable is done in the one-parameter scalar case.
Frames marked ★ generalize.

The Likelihood

Probability Versus Likelihood

Both start from the same object, the density (or mass function) $f(y \mid \theta)$. What differs is *what you hold fixed.*

Probability

$$f(y \mid \theta)$$

θ is *fixed and known*; y varies.

"If the coin is fair, how likely am I to see 7 heads in 10 tosses?"

Likelihood

$$\mathcal{L}(\theta; y) \equiv f(y \mid \theta)$$

y is *fixed* (it is your data); θ varies.

"Given that I saw 7 heads in 10 tosses, which values of θ make that outcome most probable?"

The single most important sentence today

The likelihood is *the same formula read backwards*: a function of the parameter, with the data nailed down. It is not a probability distribution over θ — it does not integrate to one, and $\mathcal{L}(\theta)$ is not "the probability that θ is true."

A First Example: Marbles in a Bag

An opaque bag contains blue and white marbles. You want the proportion θ that are blue. You draw 10 marbles with replacement and get 7 blue.

The number of blue draws is $\text{Binomial}(10, \theta)$, so

$$\mathcal{L}(\theta) = \Pr(X = 7 \mid \theta) = \binom{10}{7}\theta^7(1-\theta)^3.$$

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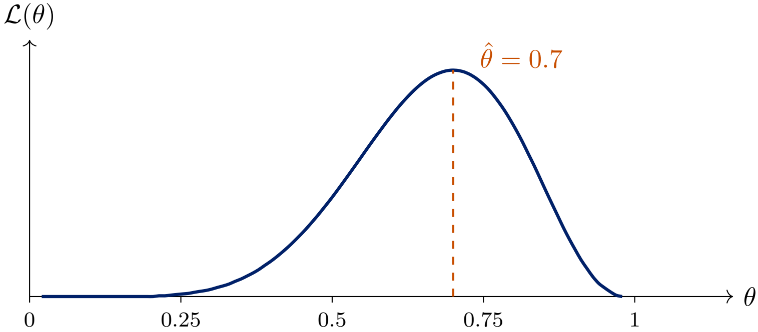
- Try $\theta = 0.3$: $\mathcal{L} = 120(0.3)^7(0.7)^3 \approx 0.0090$.
- Try $\theta = 0.5$: $\mathcal{L} = 120(0.5)^{10} \approx 0.117$.
- Try $\theta = 0.7$: $\mathcal{L} = 120(0.7)^7(0.3)^3 \approx 0.267$.

Among these, $\theta = 0.7$ makes the data we actually saw most probable. The MLE is the value that does best among *all* $\theta \in [0, 1]$.

The Likelihood, Plotted



- The height at any θ is the probability of *these* data if that θ were true.
- The peak is the MLE. The width of the peak will turn out to be our measure of uncertainty — that is the whole of Section 4.

From One Observation to Many

With n observations we need the *joint* density $f(y_1, \ldots, y_n \mid \theta)$. If the observations are independent given θ, it factors:

$$\mathcal{L}(\theta; y_1, \ldots, y_n) = \prod_{i=1}^{n} f(y_i \mid \theta).$$

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Products of many small numbers underflow and are painful to differentiate, so we take logs:

$$\boxed{\ell(\theta) \equiv \log \mathcal{L}(\theta) = \sum_{i=1}^{n} \log f(y_i \mid \theta)}$$

- log is strictly increasing, so it does not move the maximum.
- A product becomes a sum, so the derivative is a sum — which is why every MLE formula looks like a sample average.

Worked Example 1: Bernoulli

Let $Y_i \in \{0, 1\}$ with $\Pr(Y_i = 1) = \theta$. The mass function written as a single formula:

$$f(y_i \mid \theta) = \theta^{y_i}(1-\theta)^{1-y_i}.$$

(Check: at $y_i = 1$ it gives θ; at $y_i = 0$ it gives $1 - \theta$.)

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum_i y_i}(1-\theta)^{n-\sum_i y_i},$$

$$\ell(\theta) = \Big(\sum_i y_i\Big)\log\theta + \Big(n - \sum_i y_i\Big)\log(1-\theta).$$

Notice that the data enter only through $\sum_i y_i$: it is a *sufficient statistic*.

Worked Example 2: Poisson

Let Y_i be a count with $\Pr(Y_i = y \mid \theta) = \dfrac{\theta^y e^{-\theta}}{y!}$.

$$\ell(\theta) = \sum_{i=1}^{n} \log\left(\frac{\theta^{y_i} e^{-\theta}}{y_i!}\right) = \sum_{i=1}^{n} \Big(y_i \log \theta - \theta - \log(y_i!)\Big)$$

$$= \log(\theta) \sum_i y_i \; - \; n\theta \; - \; \sum_i \log(y_i!).$$

- The last term does not involve θ at all. It is an *additive constant* and can be dropped for maximization — but not if you want to compare log-likelihood *values* across models.
- Again the data enter only through $\sum_i y_i$.

Worked Example 3: Normal

Let $Y_i \sim \mathcal{N}(\mu, \sigma^2)$, two unknown parameters.

$$f(y_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mu)^2}{2\sigma^2}\right),$$

so

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2.$$

- Look at the last term: for fixed σ^2, maximizing over μ means minimizing $\sum_i (y_i - \mu)^2$. Least squares falls out of the normal likelihood, not the other way round.
- Replacing μ with $X_i'\beta$ gives the linear regression model.

Finding the MLE

The Score Function

Definition

The score is the derivative of the log-likelihood with respect to the parameter:

$$S(\theta) \equiv \frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial \log f(y_i \mid \theta)}{\partial \theta}.$$

- The MLE $\hat{\theta}$ solves $S(\hat{\theta}) = 0$ — the first-order condition, exactly as the normal equations were the FOC for least squares.
- Second-order condition: $\partial^2 \ell / \partial \theta^2 < 0$ at $\hat{\theta}$, so that we have a maximum rather than a minimum.
- A key population fact: $E[S(\theta_0)] = 0$ at the true parameter. The score is a *mean-zero* random variable — which is why the CLT will apply to it.

Bernoulli: Deriving $\hat{\theta}$

$$\ell(\theta) = \Big(\sum_i y_i\Big) \log \theta + \Big(n - \sum_i y_i\Big) \log(1-\theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta} \overset{!}{=} 0.$$

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$$\ell(\theta) = \Big(\textstyle\sum_i y_i\Big)\log\theta + \Big(n - \textstyle\sum_i y_i\Big)\log(1-\theta)$$

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Multiply through by $\theta(1-\theta)$:

$$(1-\theta)\sum_i y_i - \theta\Big(n - \sum_i y_i\Big) = 0$$

$$\sum_i y_i - \theta\sum_i y_i - n\theta + \theta\sum_i y_i = 0$$

$$\boxed{\hat{\theta} = \frac{1}{n}\sum_i y_i = \bar{y}}$$

The MLE of a probability is the sample proportion — derived, not assumed.

Poisson: Deriving $\hat{\theta}$

$$\ell(\theta) = \log(\theta)\sum_i y_i - n\theta - \sum_i \log(y_i!)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - n \overset{!}{=} 0 \qquad \Longrightarrow \qquad \boxed{\hat{\theta} = \bar{y}}$$

Check the second-order condition:

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} < 0,$$

so we really have a maximum.

Both examples give $\bar{y}$, which is not a coincidence: for these distributions the mean *is* the parameter.

Normal: Deriving $\hat{\mu}$ and $\hat{\sigma}^2$

Two parameters, so two first-order conditions.

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (y_i - \mu) \stackrel{!}{=} 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{y}.$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (y_i - \mu)^2 \stackrel{!}{=} 0 \quad \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (y_i - \hat{\mu})^2.$$

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The MLE of σ^2 is biased

It divides by n, not $n-1$. So $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2 < \sigma^2$.

This is our first warning: maximum likelihood does not promise unbiasedness. What it promises is good behaviour as n grows.

When There Is No Closed Form

Most likelihoods you will meet — logit, probit, negative binomial, anything with covariates — have no algebraic solution to $S(\theta) = 0$. We search numerically.

Newton–Raphson

Start at $\theta^{(0)}$ and iterate

$$\theta^{(t+1)} = \theta^{(t)} - \left[\left. \frac{\partial^2 \ell}{\partial \theta^2} \right|_{\theta^{(t)}} \right]^{-1} S(\theta^{(t)}),$$

i.e. take a step whose size is the score divided by the curvature. Steep curvature $\Rightarrow$ small confident steps; flat $\Rightarrow$ large steps.

Variants you will meet: **BFGS** (approximates the second derivative), **Fisher scoring** (uses the expected rather than observed second derivative — this is what `glm()` does, as iteratively reweighted least squares).

Practical Matters in `R`

- `optim()` minimizes. MLE is a maximization problem, so you pass it the *negative* log-likelihood.
- Supply `par`, the starting values. Poor starting values are the most common cause of failed convergence.
- Always check the convergence code (`$convergence == 0`) — a returned number is not the same as a converged number.
- Ask for `hessian = TRUE`; you need it for standard errors. Remember the sign: if you minimized $-\ell$, then the returned Hessian is $-\partial^2\ell/\partial\theta\partial\theta'$, which is already the observed information.
- Try several starting values. If they land in different places, your likelihood is multimodal or your model is not identified.

All of this is Thursday's lab.

Identification

Definition

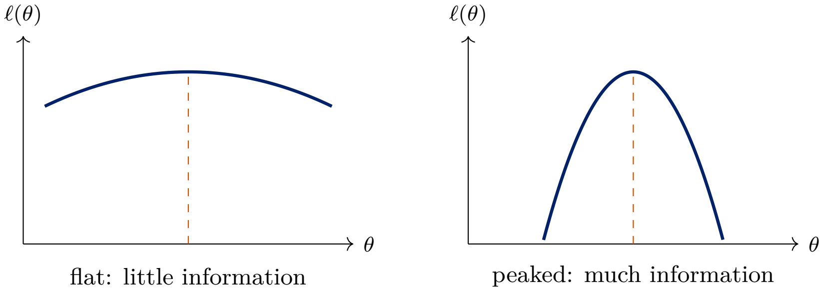
θ is identified if distinct parameter values imply distinct distributions of the data: $\theta^* \neq \theta \Rightarrow f(\cdot \mid \theta^*) \neq f(\cdot \mid \theta)$.

Non-identification means the likelihood is *flat* along some direction: the data cannot distinguish among a set of parameter values.

- **Classic example**: $Y_i \sim \mathcal{N}(\theta_1 - \theta_2,\, 1)$. Only the *difference* is identified. Optimizers started at different points will return different $(\hat{\theta}_1, \hat{\theta}_2)$ lying along a ridge, all with the same log-likelihood.
- This is the likelihood analogue of perfect collinearity in OLS — and it has the same symptom: a singular Hessian, so no standard errors.
- Diagnosis: multiple start values, and inspecting the Hessian's rank.

Information

Curvature Is Information



Both log-likelihoods peak at the same $\hat{\theta}$. But on the left, many nearby values of θ fit almost as well — we are uncertain. On the right, moving away from $\hat{\theta}$ costs a lot of fit — we are confident.

The second derivative measures exactly this.

Fisher Information

Definition (one parameter)

$$\mathcal{I}(\theta) \equiv -E\left[\frac{\partial^2 \ell(\theta)}{\partial\theta^2}\right] = E\left[\left(\frac{\partial \ell(\theta)}{\partial\theta}\right)^2\right] = \mathrm{Var}\left(S(\theta)\right).$$

The second equality is the information equality — it holds when the model is correctly specified, and it will matter a great deal in a moment.

- The minus sign is there because the log-likelihood is concave at a maximum, so the second derivative is negative.
- Information is *additive over independent observations*: with n i.i.d. observations, $\mathcal{I}_n(\theta) = n\,\mathcal{I}_1(\theta)$. This is the source of the familiar $1/n$ in variances.
- Two estimators are available: the *expected* information $\mathcal{I}(\hat{\theta})$ and the *observed* information $-\partial^2\ell/\partial\theta^2$ evaluated at $\hat{\theta}$. They are asymptotically equivalent under correct specification, and software typically reports the observed version.

The Variance of the MLE

$$\boxed{\widehat{\mathrm{Var}}(\hat{\theta}) = \mathcal{I}(\hat{\theta})^{-1} = \left[-\frac{\partial^2 \ell}{\partial \theta^2}\bigg|_{\hat{\theta}} \right]^{-1}}$$

Read it as a sentence: *the variance of the estimate is the reciprocal of the curvature of the log-likelihood at its peak.*

- Sharper peak ⇒ larger information ⇒ smaller variance.
- With several parameters ★, $\mathcal{I}$ is a matrix and $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) = \mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}$; the standard errors are the square roots of its diagonal.
- This is why `optim(..., hessian = TRUE)` is not optional.

Worked: Bernoulli Information

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta},$$

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} - \frac{n - \sum_i y_i}{(1-\theta)^2}.$$

Taking expectations, using $E[\sum_i y_i] = n\theta$:

$$\mathcal{I}(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta} + \frac{n}{1-\theta} = \frac{n}{\theta(1-\theta)}.$$

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$$\mathrm{Var}(\hat{\theta}) = \mathcal{I}(\theta)^{-1} = \frac{\theta(1-\theta)}{n}. \quad \checkmark$$

Exactly the variance of a sample proportion you already know — now *derived* from the shape of the likelihood.

Worked: Poisson Information

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} \quad \Longrightarrow \quad \mathcal{I}(\theta) = \frac{E[\sum_i y_i]}{\theta^2} = \frac{n\theta}{\theta^2} = \frac{n}{\theta},$$

so

$$\mathrm{Var}(\hat{\theta}) = \frac{\theta}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\theta}) = \frac{\bar{y}}{n}.$$

Sanity check by direct calculation: $\hat{\theta} = \bar{y}$ and $\mathrm{Var}(\bar{Y}) = \mathrm{Var}(Y)/n = \theta/n$ because the Poisson mean equals its variance. ✓

Two routes, same answer. When you can compute the variance directly, do so — it is a free check on your information calculation.

Properties

The Four Properties

Under regularity conditions, as $n \to \infty$:

1. **Consistency**: $\hat{\theta} \xrightarrow{p} \theta_0$.
2. **Asymptotic normality**: $\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta_0, \mathcal{I}(\theta_0)^{-1})$.
3. **Asymptotic efficiency**: among *regular* estimators, and when the model is correctly specified, none has a smaller asymptotic variance — the MLE attains the Cramér–Rao lower bound. This is much stronger than Gauss–Markov, which only ranked *linear* unbiased estimators.
4. **Invariance**: if $\hat{\theta}$ is the MLE of θ, then $c(\hat{\theta})$ is the MLE of $c(\theta)$ for any continuous $c(\cdot)$. Wonderfully convenient: the MLE of the odds is the odds of the MLE.

The first three are asymptotic — none of them says anything about $n = 40$. Invariance, by contrast, holds exactly in any sample.

Why Consistency and Normality Hold ★

The argument is a first-order Taylor expansion of the score around θ_0:

$$0 = S(\hat{\theta}) \approx S(\theta_0) + \frac{\partial S(\theta_0)}{\partial \theta}(\hat{\theta} - \theta_0) \implies \sqrt{n}(\hat{\theta} - \theta_0) \approx \frac{\frac{1}{\sqrt{n}} S(\theta_0)}{-\frac{1}{n}\partial S(\theta_0)/\partial\theta}.$$

- The **denominator** is a sample average of second derivatives; by the LLN it converges to $\mathcal{I}_1(\theta_0)$.
- The **numerator** is a normalized sum of the i.i.d. mean-zero scores; by the CLT it is asymptotically $\mathcal{N}(0, \mathcal{I}_1(\theta_0))$.
- Combining: $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta_0)^{-1})$.

This is structurally identical to the OLS argument in Week 2: a sample average in the numerator, a sample average in the denominator, LLN below and CLT above.

Regularity Conditions — and When They Fail

The theory needs: a correctly specified and identified model; a parameter space in which θ_0 is an *interior* point; a support for y that does not depend on θ; and enough smoothness to differentiate twice under the integral.

Failures you will actually meet

- **Boundary**: testing whether a variance component is zero. The usual χ^2 reference distribution is wrong.
- **Support depends on θ**: $Y_i \sim U[0, \theta]$ gives $\hat{\theta} = \max_i Y_i$, which is *always* below θ — badly biased and not asymptotically normal.
- **Separation** in logit: the MLE runs off to infinity. Week 4.
- **Too many parameters relative to n**: incidental parameters.

Maximum Likelihood Is Not Unbiased

- We already saw it: $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_i (y_i - \bar{y})^2$ has $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$.
- $Y_i \sim U[0, \theta]$: the MLE is $\max_i Y_i$, which is below θ with probability one.
- And unbiasedness does not survive transformation: even if $\hat{\theta}$ is unbiased for θ, $c(\hat{\theta})$ is generally biased for $c(\theta)$. (Invariance holds for the *MLE property*, not for unbiasedness.)

The trade we are making

Give up exact unbiasedness in finite samples; get consistency, asymptotic efficiency, and a completely general recipe that works for any probability model you can write down.

When the Model Is Wrong: the Sandwich Returns ★

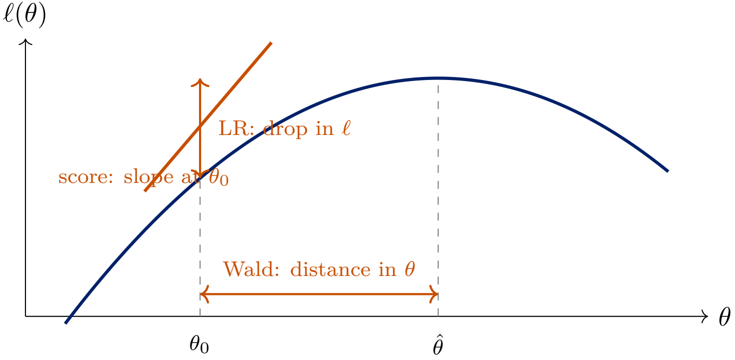
The information equality $-E[\partial^2 \ell / \partial \theta^2] = E[S(\theta)^2]$ holds only if the model is correctly specified. When it is not, the two sides differ and

$$\mathrm{Var}(\hat{\theta}) \approx \underbrace{\left[-E\partial^2 \ell\right]^{-1}}_{\text{bread}} \underbrace{E\left[S(\theta)S(\theta)'\right]}_{\text{meat}} \underbrace{\left[-E\partial^2 \ell\right]^{-1}}_{\text{bread}}.$$

- Exactly the sandwich from Week 2, with the score in place of $X_i \epsilon_i$. It is called the *quasi*- or *pseudo-MLE* variance, or the Huber–White variance for likelihood models.
- In R: `sandwich::vcovHC()` works on `glm` objects too, and `vcovCL()` clusters them.
- The estimator $\hat{\theta}$ then converges to the parameter of the *best-approximating* model in the Kullback–Leibler sense — the likelihood analogue of "best linear approximation to the CEF."

Testing

Three Ways to Measure the Same Distance



All three ask: *is* θ_0 *far from* $\hat{\theta}$*?* They are asymptotically equivalent but can differ in finite samples.

The Likelihood Ratio Test

Compare the maximized log-likelihood with and without the restriction:

$$LR = -2\big[\ell(\hat{\theta}_{\text{restricted}}) - \ell(\hat{\theta}_{\text{unrestricted}})\big] \xrightarrow{d} \chi^2_q,$$

where q is the number of restrictions.

- Logic: if the restriction is true, imposing it should not cost much fit.
- The restricted likelihood is always weakly *lower*, so $LR \geq 0$.
- **Requires estimating both models.** Usually easy, and generally the best-behaved of the three in finite samples — prefer it when you can.
- R: `lmtest::lrtest(m0, m1)`, or `anova(m0, m1, test = "Chisq")` for `glm`.
- The models must be nested and fitted on the same observations. Silently dropping different rows to missingness invalidates the test.

The Wald Test

Measure the horizontal distance, scaled by the estimated standard error:

$$W = \frac{(\hat{\theta} - \theta_0)^2}{\widehat{\mathrm{Var}}(\hat{\theta})} \xrightarrow{d} \chi^2_1, \qquad \text{or} \quad z = \frac{\hat{\theta} - \theta_0}{\widehat{\mathrm{SE}}(\hat{\theta})}.$$

- This is the t-ratio you already read off every regression table.
- **Requires only the unrestricted model** — convenient when the restricted model is hard to fit.
- Multi-parameter version ★: $W = (\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})'[\boldsymbol{R}\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}})\boldsymbol{R}']^{-1}(\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})$ — identical in form to Week 2.
- Caution: the Wald test is not invariant to how you parameterize the restriction. Testing $\theta = 1$ and $\log\theta = 0$ can give different answers. The LR test does not have this problem.

The Score (Lagrange Multiplier) Test

Measure the slope of the log-likelihood *at the null*:

$$LM = \frac{S(\theta_0)^2}{\mathcal{I}(\theta_0)} \xrightarrow{d} \chi^2_q.$$

- Logic: if θ_0 were the maximum, the slope there would be zero. A steep slope is evidence against it.
- **Requires only the restricted model** — useful when the unrestricted model is expensive to fit, e.g. testing for overdispersion without fitting the negative binomial.
- Many familiar specification tests are score tests in disguise, including Breusch–Pagan for heteroskedasticity.
- **Which to use?** Asymptotically the three are identical. In finite samples: LR if you can fit both models, Wald if only the unrestricted is available, score if only the restricted is.

Model Comparison Beyond Nested Tests

LR, Wald and score all require *nested* models. For non-nested comparison:

$$AIC = -2\ell(\hat{\theta}) + 2k, \qquad BIC = -2\ell(\hat{\theta}) + k \log n.$$

- Both penalize fit by complexity; lower is better.
- BIC penalizes more heavily once $n > 7$, and increasingly so with n. BIC targets the "true" model if it is in the candidate set; AIC targets predictive accuracy.
- Neither is a hypothesis test; there is no p-value and no threshold.
- Comparable only across models fitted on *exactly the same* data with the same outcome scale.

We return to model selection properly in Week 12, where cross-validation replaces these analytic penalties.

Where This Leaves Us

- A likelihood is the data's probability read as a function of the parameter. The MLE maximizes it; the score is the derivative you set to zero.
- The *curvature* of the log-likelihood is information, and its inverse is the variance. Everything about precision follows from the shape of the peak.
- The MLE is consistent, asymptotically normal, and efficient — but not unbiased, and all of the guarantees are asymptotic.
- LR, Wald, and score are three views of the same question.

Next

Thursday: code a likelihood and maximize it numerically in `R`.
Week 4: put covariates inside the likelihood — the generalized linear model, and the limited dependent variable models you will actually use.

References

- Pawitan, Yudi. 2013. *In All Likelihood: Statistical Modeling and Inference Using Likelihood.* Oxford University Press. [Ch. 2 and 4 required; Ch. 3 additional]
- Casella, George and Roger L. Berger. 2002. *Statistical Inference*, 2nd ed. Duxbury. [Ch. 7 and 10, for a fuller treatment]
- King, Gary. 1998. *Unifying Political Methodology: The Likelihood Theory of Statistical Inference.* University of Michigan Press.
- White, Halbert. 1982. "Maximum Likelihood Estimation of Misspecified Models." *Econometrica* 50(1): 1–25. [the quasi-MLE sandwich]